

Discovery Philosophy Engineering. Making Philosophical Claims Experimentally Interrogable

Abstract

Philosophy is rich in claims about which distinctions matter, which descriptions are equivalent, what counts as rational or objective, and which structures are fundamental. Many such claims stay insulated from empirical pressure because they are stated at a level too informal to generate a discriminating test. That invariance under admissible transformation marks what is real rather than how we happened to describe it is an old idea with four developed statements already, in the study of what a transformation group leaves fixed, in robustness as agreement across independent means of detection, in invariance under intervention as the mark of a causal relation, and in the structure that survives the replacement of one theory by another. This paper does not propose a fifth. It proposes that the idea be made to cost something, by compiling a philosophical claim into an executable invariance test, declaring the admissible class in advance as a versioned artifact, reducing each failure to the smallest transformation that produces it, naming the component that absorbs the change, and recording the revision that follows. The result is a discipline we call Discovery Philosophy Engineering, in which an implementation is not the endpoint of philosophical analysis but an instrument for experimenting on it. We define an invariance envelope as the family of transformations under which a claim survives, a boundary witness as the smallest admissible transformation that breaks it, and a comparative notion of fundamentality that ranks two descriptions of comparable adequacy by their envelopes. We then report the discipline running. A campaign graded four of its own theory's predictions through registrations sealed before the data were read, returning one pass whose interesting number is the half of the population that violates the law, two indeterminate verdicts whose reduction located where a theorem stops applying, and one pass in which two of three declared budgets behaved as the theory said and the third did not. A discovery line asked one question nine times, failed three of the nine by clauses written in advance, and ended with one law, one declared scope, and two recorded boundaries. The check the discipline prescribes then found a defect in its own authors' records that reading them had not. What is new here is not a criterion of invariance. It is the accounting that lets somebody else test one.

Keywords: invariance, robustness, objectivity, scientific discovery, representation independence, theory revision, fundamentality, preregistration, computational philosophy

1 Introduction

Philosophical inquiry repeatedly asks questions of one form. Which distinctions are real? Which descriptions are different presentations of the same case? Which aspects of a judgment are objective, and which depend on a standpoint? What is preserved when a problem is translated, reparameterized, rescaled, or embedded in a richer representation? And when two theories disagree, which disagreement reflects the world and which reflects the language in which the world was described?

These are questions about invariance, and philosophy has answered them in that vocabulary for a long time. Section 2 sets out four developed answers, because the argument here depends on

conceding how much of its central idea is already held. What this paper adds is not another account of why invariance matters. It is a way of making a particular invariance claim cost something to assert.

The difficulty is not that philosophers lack a criterion. It is that a criterion stated in prose leaves three things in the claimant's hands and shows the reader none of them. The class of transformations said to be irrelevant is chosen by the theorist, and a class chosen after the result is a class fitted to it. A failure is reported as a failure, without the smaller failure inside it that would say which assumption gave way. A commitment revised in the light of a counterexample is revised silently, so the record cannot separate a theory that learned from a theory that was never wrong. None of these is a defect of the invariance idea. Each is a defect of the accounting around it.

Experimental philosophy has subjected philosophical judgments to empirical study and found both stability and fragility in methods of cases (Knobe and Nichols, 2024; Irikefe, 2025). Computational philosophy uses mechanized methods to extend philosophical analysis (Thagard, 2024). Formal philosophy turns conceptual commitments into mathematical structures. Automated scientific discovery searches data for compact laws, increasingly through symbolic regression and closed-loop systems (Makke and Chawla, 2024; Kramer et al., 2026). Each supplies a necessary piece. None supplies a discipline for taking one philosophical proposition, compiling it into invariance and boundary claims, declaring in advance what would refute it, attacking it, and recording what the attack did to it in a form a third party can check.

This paper proposes that discipline, which we call Discovery Philosophy Engineering. It extends Philosophy Engineering, a method for operationalizing philosophical commitments through declared equivalences, invariance requirements, enforcement mechanisms, and machine-checkable artifacts (Anonymous, 2026c). The extension closes the loop. Rather than moving only from a philosophical principle to a formal constraint to an implementation, the implementation is turned back on the principle as an instrument. If a proposition can be formalized well enough to implement, the implementation usually permits experiments on it that argument alone cannot perform.

The contribution is methodological. Many philosophical questions are conceptual, normative, or modal in ways no test settles, and the discipline claims nothing about those. It applies where a philosophical claim entails observable invariance, representation, discrimination, or boundary structure, and its aim is to sharpen those entailments until success and failure are both informative.

2 Invariance has been asked to do this work before

The thought that what survives admissible transformation is a better candidate for the real than what does not is old, and it has at least four developed statements. A proposal in this area earns attention only by saying what it adds to them.

Geometry as the study of what a group leaves fixed. The Erlangen reading classifies geometries by the transformation groups whose invariants they study (Klein, 1893), and the extension of that reading to physical law is Weyl (1952). This is the origin of the move and remains its clearest case. What it supplies is the form of the question. What it does not supply is guidance when the admissible group is not handed over by the subject matter, which is the normal situation outside mathematics and physics.

Objectivity as invariance. Nozick (2001) makes invariance under admissible transformations the account of objectivity itself, so that an objective fact is one holding across the transformations that do not matter. This is the closest existing statement of the present thesis, and Definition 3 below is a near relative of it. Nozick offers an analysis of a concept. He does not offer a procedure

for deciding a particular case, and the question of who fixes the admissible transformations is raised rather than answered.

Robustness as agreement across independent means of detection. Levins (1966) defended robustness analysis in model building and Wimsatt (1981) generalized it, using the word robust for exactly what is invariant across independent means of determination. This is the structure of the invariance envelope of Section 4.3, reached three decades earlier and from the practice of working scientists rather than from geometry. The difference is bookkeeping rather than concept. Robustness analysis reports that a result held across several independent routes. It does not generally declare the set of routes in advance, does not require failures to be reduced, and has no standard form in which a reader can see which routes were tried and rejected. The literature knows this, and the charge that robustness can be manufactured by choosing agreeable models is the standing objection to it.

Invariance under intervention. Woodward (2003) makes invariance under a range of interventions the mark separating a causal generalization from an accidental one, and treats invariance explicitly as a matter of degree and range rather than as a predicate. The degree and range treatment is the move made here with the envelope, and Woodward’s account is the more developed of the two as philosophy. What it is not is a procedure with an audit trail, since the range over which a relation is invariant is established by argument and by experiment reported in the ordinary way.

Structure as what survives theory change. Structural realism holds that what carries across a scientific revolution is mathematical structure rather than ontology (Worrall, 1989). Replace the transformation from one theory to its successor by a transformation from one representation to another and the resemblance is close. Ismael and van Fraassen (2003) sharpen the related point that a symmetry marks superfluous theoretical structure, which is the philosophical form of the factorization criterion stated as Proposition 1 below. Their argument and that proposition say one thing in two vocabularies.

The concession is therefore large. The criterion is not new, the treatment of invariance as a matter of degree is not new, the use of a symmetry to identify structure that can be quotiented away is not new, and the identification of objectivity with invariance is not new. Five of this paper’s central ideas have owners.

What none of them supplies is an account under which an invariance claim can be checked by somebody who does not trust the claimant. In each, the admissible class is settled inside the argument that uses it. That is not a criticism of the authors, because prose was the medium they had. It is a description of what is left to be added. The proposal here is that the three things a prose statement leaves implicit, the class, the reduction of the failure, and the revision, become artifacts declared before the test, fixed by content hash, and refused by a checker when absent. Mayo (2018) supplies the notion that makes this worth doing, since a test is severe when the claim would probably have failed it had the claim been false, and severity is exactly what a class chosen after the fact destroys.

The novelty claimed here is accordingly narrow. It is not a new criterion of invariance. It is the accounting that lets an old one be operated by more than one person.

3 A claim becomes an instrument when its equivalences are declared

Philosophy Engineering was introduced as the systematic translation of epistemic, ethical, and logical commitments into computationally enforceable constraints on autonomous systems (Anonymous, 2026c). Its procedure has four steps. Declare a domain-specific equivalence relation over

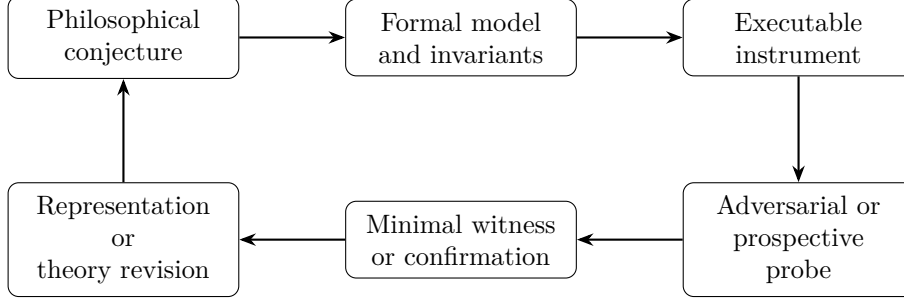


Figure 1: The loop the discipline runs. The arrow that separates it from ordinary hypothesis testing is the one from witness to revision, since a counterexample that has been reduced says which assumption must give way and an unreduced one does not.

representations. Declare what must remain invariant. Implement canonicalization, quotienting, equivariance, or another enforcement mechanism. Package transformation trials and witnesses as machine-checkable artifacts. Its central example is the Epistemic Invariance Principle, which holds that a judgment procedure should not change its conclusion under transformations declared to preserve task-relevant meaning.

Discovery Philosophy Engineering generalizes this in the reverse direction. A philosophical claim becomes a source of formal commitments that a transformation grammar and an observation process can interrogate. The engineering artifact stops being the endpoint of analysis and becomes an instrument for experimenting on it.

Principle 1 (Discovery Philosophy Engineering). *A philosophical conjecture is a candidate for Discovery Philosophy Engineering when it can be translated into at least one non-trivial formal constraint whose satisfaction or violation is observable under a declared family of transformations, interventions, or representation changes.*

Non-triviality is essential. A constant judgment rule is invariant under every transformation and has learned nothing. Philosophy Engineering therefore pairs invariance with discriminative adequacy, so that irrelevant differences must not matter and relevant differences must be allowed to matter, and the present discipline inherits that dual obligation. Figure 1 shows the loop.

4 Formal architecture

4.1 Representations, transformations, and judgments

Let S denote a space of world states or cases, X a representation space, and $\sigma : S \rightarrow X$ an encoding map. Let Y be an output space, and let a judgment or theory-induced prediction be

$$J : X \rightarrow \mathcal{P}(Y), \tag{1}$$

where $\mathcal{P}(Y)$ is the space of distributions over Y and a deterministic judgment is a degenerate distribution. Let $\mathcal{T}$ be a set of transformations $\tau : X \rightarrow X$ declared to preserve some target-relevant structure.

The declaration matters, and it is where this architecture departs from the accounts of Section 2. No method discovers relevance without assumptions. An admissible transformation is not anything that can be done to the data. It is a transformation licensed by the philosophical claim

under test. Unit conversion, variable renaming, permutation of independent premises, paraphrase preserving formal denotation, change of basis, population transfer, and stake rescaling are different transformations with different semantics, and a claim surviving one says nothing about another.

Definition 1 (Invariance claim). *Given an output equivalence relation $\sim_Y$, the judgment J is invariant under $\mathcal{T}$ if*

$$J(x) \sim_Y J(\tau x) \quad \forall x \in X, \tau \in \langle \mathcal{T} \rangle, \quad (2)$$

where $\langle \mathcal{T} \rangle$ is the closure generated by $\mathcal{T}$ when composition is meaningful.

This is the formal core of the Epistemic Invariance Principle (Anonymous, 2026c). Read as a compliance property it is a specification. Read as this paper reads it, it is an empirically attackable consequence of the philosophical commitment that the transformations in $\mathcal{T}$ do not matter.

4.2 A theory calling a difference representational should factor through the quotient

Let $x \sim_{\mathcal{T}} x'$ when two representations lie in the same orbit under the admissible transformations. The quotient $X / \sim_{\mathcal{T}}$ identifies every description the theory says is the same case.

Proposition 1 (Factorization criterion). *If J is constant on every $\sim_{\mathcal{T}}$ equivalence class, then there exists a well-defined map $\bar{J} : X / \sim_{\mathcal{T}} \rightarrow \mathcal{P}(Y)$ such that $J = \bar{J} \circ q$, where $q : X \rightarrow X / \sim_{\mathcal{T}}$ is the quotient map.*

Proof. Define $\bar{J}([x]) = J(x)$. Class constancy makes the definition independent of the representative chosen. Then $\bar{J}(q(x)) = J(x)$ by construction. $\square$

The mathematics is elementary and the reading carries the weight. A theory calling certain differences merely representational is committed to a judgment factoring through the quotient that removes them, and a failure to factor locates a representational dependence the theory did not admit to having. This is the content of the argument that a symmetry marks superfluous theoretical structure (Ismael and van Fraassen, 2003), stated so a program can test it on a particular case.

4.3 Invariance is an envelope rather than a predicate

Pass and fail are too coarse for discovery. A theory may be invariant under unit changes and not under scale changes, stable across paraphrases and not across languages, transferable to low-stakes contexts and not to consequential ones.

Definition 2 (Invariance envelope). *For a theory or judgment procedure J and a universe U of candidate transformations, the invariance envelope is*

$$\mathcal{E}(J) = \{ \tau \in U : J(x) \sim_Y J(\tau x) \text{ over the declared test domain} \}. \quad (3)$$

The test domain must be reported, because an empirical envelope is always relative to sampled cases, measurement tolerance, and statistical power. The formal object is an idealization and empirical work estimates it. Treating invariance as a matter of degree and range is Woodward's move (2003), and the set of routes across which a result holds is Wimsatt's (1981). The envelope adds one requirement to both, that U be written down before the test rather than recovered from it afterwards.

Definition 3 (Comparative fundamentality). *Let J_1 and J_2 be candidate descriptions with comparable empirical adequacy and without compensating increases in unconstrained complexity. Then J_1 is more invariantly fundamental than J_2 relative to U if $\mathcal{E}(J_2) \subseteq \mathcal{E}(J_1)$.*

This is comparative and local by design. It does not identify metaphysical fundamentality with invariance, and it is not offered as a contribution to the metaphysics of fundamentality, where the relations at issue are grounding and ontological priority rather than survival under redescription. It says only that when two descriptions account for the same phenomena, the one surviving more meaning-preserving changes has discarded more representational contingency. A reader who holds Nozick’s (2001) account of objectivity may read the definition as that account restricted to a declared U and made decidable for a particular pair.

Remark 1. The complexity qualification blocks a vacuous victory in which invariance is bought by adding a correction for every transformation. The discipline therefore pairs envelope breadth with parsimony, preregistration, and prospective prediction. Without the qualification, comparative fundamentality would reward exactly the ad hoc manoeuvre the robustness literature identifies as its own main risk.

4.4 A failure says more when it is reduced to its smallest witness

A radar return is useful because it localizes. The analogue for theories is a counterexample small enough to name a cause.

Definition 4 (Boundary witness). *A boundary witness for an invariance claim is a transformation or intervention τ^* such that $J(x) \not\prec_Y J(\tau^*x)$ for some x , while no strictly simpler admissible predecessor under a declared complexity ordering produces the violation.*

For a finite transformation grammar, breadth-first search identifies a minimal violating composition under word length (Anonymous, 2026c). The move is standard in software debugging, where delta debugging narrows a failing input to a minimal failure-inducing core (Zeller and Hildebrandt, 2002), and the methodological content is identical. A reduced failure names a component and an unreduced one does not. For continuous parameters the boundary is estimated as

$$t^* = \inf\{t : D(J(x), J(T_t x)) > \epsilon\}, \quad (4)$$

with D an output discrepancy and ϵ a tolerance fixed in advance. A failed invariance test then measures where a degree of freedom supposed to be irrelevant starts to matter.

5 The protocol

The discipline organizes work as a sequence of increasingly severe tests, severe in Mayo’s sense (2018). The objective is not to maximize the phenomena a theory can be made to fit after the fact. It is to find the smallest structure that still predicts after everything the theory says should not matter has been changed.

5.1 Formalize the commitment

Replace broad language, such as objective, rational, same case, irrelevant framing, and fundamental, with explicit mathematical commitments. These may be equivalence relations, group actions, order restrictions, metric identities, conservation conditions, commutative diagrams, or conditional

independences. Formalization must separate four statuses, which are definitions, modelling assumptions, derived propositions, and empirical hypotheses. The common error in interdisciplinary work is treating a convenient representation as an observed law, and the separation prevents it.

5.2 Derive consequences that divide rivals

A good probe is not one under which the theory could fail. It is one under which competing theories diverge. Candidate consequences should be ranked by expected discriminatory value, and invariance where rivals predict sensitivity, sign reversals, boundary locations, low-rank constraints, monotonicity, and forbidden orderings are the useful kinds. The strongest tests are often consequences the theorist would not have chosen, and they are valuable because the formal structure forces them.

5.3 Search for low-complexity structure

The discovery engine may be human reasoning, symbolic regression, equation discovery, structure learning, or a hybrid. Symbolic regression is attractive because it searches for concise relations rather than only for predictive black boxes (Makke and Chawla, 2024), recent work applies it to interpretable model discovery under physical constraints (Hou et al., 2024), and broader automated-science systems combine representation learning, equation discovery, experiment selection, and closed-loop autonomy (Kramer et al., 2026). A search objective can be written

$$\min_{F \in \mathcal{H}} L(F) + \lambda C(F) + \mu I(F), \quad (5)$$

with L a predictive loss, C a structural complexity, and I a penalty on instability under transformations the programme has declared irrelevant. This is not proposed as a unique objective. It states the preference for descriptions that are accurate, simple, and invariant together.

5.4 Attack the representation

Subject a candidate law to transformations preserving information while changing coordinates or presentation. Basis rotations, unit changes, equivalent encodings, feature permutations, redundant embeddings, alternative encoders, and controlled compressions are the usual instruments. Structure disappearing under an information-preserving change of coordinates may belong to the representation rather than to the phenomenon. Contemporary symmetry discovery does the corresponding technical work, enforcing known symmetries, searching for unknown ones, and breaking candidate symmetries only where data require it (Otto et al., 2025), and causal representation learning has been reframed so identifiability runs through preserved invariances rather than one fixed formalism (Yao et al., 2025). The philosophical reading is that a distinction should be kept only if the world forces us to keep it.

5.5 Freeze, then probe what was not used to build it

After discovery, freeze parameters and transformation rules as far as possible and evaluate on unseen cases, populations, scales, or domains. This is where a suggestive reduction becomes evidence of transferable structure. Table 1 states the ladder.

Level	Question
Fit	Can the structure describe the observed cases?
Perturbation	Does it survive local parameter and data perturbation?
Reparameterization	Does it survive information-preserving changes of coordinates?
Holdout	Does it predict unseen cases from the same population?
Transfer	Does the same structure survive a different task or population?
Cross-domain	Does it organize qualitatively different phenomena?
Adversarial probe	Does it survive a deliberately chosen discriminator?
Boundary map	Where does it cease to apply, and which assumption breaks?
Replication	Can an independent group reproduce the chain?

Table 1: The ladder of probes. A result earns only the rung it has reached, and the rung a programme reports is the first thing a reader should check, since every rung above Holdout is one a well-fitted artifact can still fail.

5.6 Reduce failures rather than hide them

A null result may be uninformative. A localized contradiction is not. When an invariance fails, find the smallest witness and ask which component absorbs the change, the candidates being the representation, the metric, the constraints, the resolution budget, the dynamics, and the philosophical equivalence relation itself. This turns falsification from a verdict into cartography. The aim is not to immunize a theory against failure but to make its failures diagnostic.

6 Where this sits among existing methods

Experimental philosophy studies the distribution and stability of judgments elicited by cases, and recent work continues to debate calibration, comprehension, and robustness to error (Knobe and Nichols, 2024; Irikefe, 2025; Cova and Martinez, 2025). The present discipline shares the commitment that philosophical practice can be interrogated empirically and differs in its unit of analysis, targeting formal consequences of philosophical theories rather than primarily the psychology of philosophers or lay participants. Human judgments may be the dependent variable, and so may physical measurements, algorithmic behaviour, economic choices, or theorem-prover outcomes.

Computational philosophy uses simulation, automated reasoning, and formal modelling to instantiate and extend philosophical research (Thagard, 2024), and this is a narrower proposal inside that space, distinguished by the closed loop from conjecture to executable test to reduced witness to revision. Formal philosophy supplies the language without which the proposal would be metaphor, and the discipline adds the requirement that a formal claim with observable consequences also specify how those consequences are tested and what would count as a minimal counterexample. Automated scientific discovery has moved from equation discovery toward systems selecting experiments and learning representations (Kramer et al., 2026), and the difference here is the type of prior knowledge compiled, since the starting commitments are philosophical and part of the goal is to find which of them survive contact with the world.

7 The discipline at work

Five instances follow. The first is generic. The next three are reported briefly, each making a single point. The fifth is reported in full, because it is the only case in which the protocol of Section 5 ran

to its own declared end. The sixth is the case in which the discipline’s own check found a defect in the records of the programme that wrote it.

Every number in Sections 7.2, 7.5, and 7.6 is read from a sealed record in a public repository, cited where it appears. None is restated from a draft.

7.1 Epistemic invariance as an experimental object

The Epistemic Invariance Principle says a judgment should remain equivalent under meaning-preserving transformations (Anonymous, 2026c). In its engineering role it is a specification, so a system changing a medical recommendation because laboratory results were reordered violates a declared norm. The discovery role is different.

Suppose a class $\mathcal{T}$ is proposed as semantically irrelevant. An engineer can build the transformation suite, measure judgment discrepancy, and search for minimal witnesses. Three outcomes are possible. Invariance may hold broadly, and the claim that the transformed inputs are the same case gains operational support. Invariance may fail because the model is representation-sensitive while the philosophical equivalence remains compelling, in which case the system is defective against the norm. Or invariance may fail systematically in a way showing the proposed equivalence erased a relevant distinction, in which case the philosophy and not only the implementation should change. The third outcome is the distinctive one, and it is where engineering exposes a hidden philosophical assumption by forcing it to meet a concrete counterexample.

7.2 A campaign that graded four of its own predictions

Geometric Evaluation Theory is an evaluator-relative theory in which states and actions map into a consequence space, an evaluator supplies a metric and an ideal point, a finite budget limits the distinctions the evaluator can make, and preference and choice follow from those distinctions (Anonymous, 2026a). Its draft separates what the world supplies, the states, the actions, and the consequence map, from what the evaluator owns, the metric, the budget, and the admissible set. Every claim is labelled proved, defined, posited, or open, and predictions are graded through gates sealed before the data are read. Four of its empirical questions have run across five sealed registrations.

The first tested the hull law, a consequence of the representation theorem that no action is strictly worse than every action whose consequences surround it. The consequence map must be known, so the world was the 1972 American National Election Study, in which respondents placed candidates and parties on seven-point issue scales and rated them on feeling thermometers. The registration was sealed before any rating was read. On the registered pair of scales 489 respondents had a menu in which the law could bind, 50 percent violated it at least once, against 66 percent when each respondent’s ratings were shuffled among that respondent’s own objects, and no shuffle in 200 reached the observed rate. Five other pairs of scales gave the same ordering. The verdict by the sealed bars was a pass, and the number that matters more than the verdict is the half. The theorem is deterministic for a representable preference, so the law holds in this population as a tendency and fails as a law for half the respondents. In the vocabulary of Section 4.4 the population is a boundary witness against the deterministic reading, and what absorbs it is not yet registered, since no error model for integer placements and integer ratings was part of the seal.

The second tested the claim that the indifference threshold of the induced order is the evaluator’s resolution budget. The evaluator was a language model asked how far a value sits from a target and answering with a number, and three budgets were set independently, the decimals shown, the tokens allowed for the report, and the precision of the weights. A cell’s threshold is the separation

at which the scorer orders 90 percent of pairs correctly. On fresh pairs every graded threshold sat at 0.87 to 0.88 of the step the rendering imposed and of the resolution the report length afforded, the largest ratio 1.15 against a tolerance of 1.5 fixed before the seal, and the constant near 0.87 is where the 90 percent level of a rounding scorer falls, as a self-test on a known grid had shown. The weight precision did not move the threshold at all, eight-bit and four-bit weights leaving it where full precision put it to the third digit. Two of three declared budgets are resolution budgets and the third is not, which is the finding, and the registration could have carried only the two that worked. Two instruments tried before the one that ran are kept in the record rather than dropped. A pairwise chooser read from answer-letter logits was vacuous when the options straddled the target, because the model picks the larger number, and had a floor of 7.5 when they did not, from position bias. The reduction of that failure generalizes past the case. A pairwise chooser measures its own position bias before it measures a threshold.

The third tested the incompatibility theorem, which says several evaluators reading one rank-limited representation each pay a regret fixed by their geometries, that a shared representation best for all exists only when their leading subspaces coincide, and that least total regret is attained by the leading eigenspace of the summed geometries. The world was the grouped-query attention of a language model, with three query heads as evaluators of one key cache and shared codes of rank 2, 4, and 8 in 128 dimensions. The first sealed run coded all keys at once and found the measured attention loss 20 to 330 times the second-order prediction, so the regret formula was not tested and the verdict was indeterminate. The ordering claim survived, the leading eigenspace of the summed geometries beating every random code and the key-covariance code in 32 of 32 cases, while a head's own leading eigenspace was best for that head in fewer than half. The failure was then reduced, as Section 5.6 requires. The registered prediction and the registered measurement were not the same object, a one-key operator against an all-keys loss, and the codes sat far outside the range where the quadratic approximation holds. A second sealed run measured the object the formula predicts, one key perturbed by a small isotropic error in the discarded subspace, along a ladder of perturbation sizes. The ratio of measured to predicted loss was flat in the perturbation size up to a tenth of a whitened unit with median over cells 1.02 to 1.05, so the recovered operators predict the loss without bias, and 64 draws per cell left a 20 percent scatter that missed the registered per-triple bar, so that verdict was indeterminate as well. A seed check with fresh draws showed the scatter to be draw noise. Two errors in that registration's own error estimate are recorded in it. What the two runs establish together is the boundary of the theorem's range of validity. Inside it the regret formula and the bound follow from the recovered operators, so the only empirical content is whether the loss is quadratic with the recovered operator, which holds. Outside it the ordering survives and the private optimum does not. The perturbation ladder is equation (4) executed.

The fourth tested identification, the theorem that an evaluator's choices recover the metric up to scale and the ideal up to the metric's kernel, with the design bound that the consequences must span the space affinely. The estimator is the representation theorem's semidefinite program. Two pilots before the seal caught two wrong drafts of that estimator and showed the drafted absolute tolerances were empty, since identification from ordinal data is slow, so the bars were restated on medians relative to each evaluator's chance level. The sealed run passed in all twelve cells on a fresh seed. From a weak order on 64 consequences above the dimension, the metric and the ideal's range component were recovered to a few percent of chance with error falling at every step of the battery ladder. At or below the design bound, off the span of a battery confined to a subspace, and in the kernel of a singular metric, nothing was recovered. A semiorder withholding the fifth of pairs closest in distance identified more slowly. The negative claims, what choices cannot reveal, came out as sharp as the positive one, and each would have been a witness against the theorem or its remark had it failed.

This theory suits the discipline because it makes representation choice explicit. Under an invertible change of basis $z' = Az$ a quadratic metric transforms as

$$G' = A^{-T}GA^{-1}, \quad (6)$$

so that

$$z'^T G' z' = z^T G z, \quad (7)$$

and the evaluative distance is invariant while the coordinates are not. Which properties of evaluation survive changes of representation, encoder, dimensionality, and task is then the question the campaign asks gate by gate. The campaign file, the ledger, the sealed registrations, the run logs, and every result file are public.

7.3 A published invariance that failed on independent data

A geometric model of economic behaviour accepted by a systems journal contains both a successful reduction and an independent failure (Anonymous, 2026e). It represents alternatives in a nine-dimensional decision space and fits a diagonal Mahalanobis metric. Structural search selects social impact, virtue and identity, and epistemic status as active dimensions on the reported benchmark, and the monetary dimension is inactive in the selected low-stakes calibration.

The methodological step is not the fit but the consequence the inactive monetary coordinate forces. Under the share-normalized encoding, positive rescaling of all monetary stakes should leave predicted offers and rejections unchanged, which is a sharp stake-scaling invariance. Tested on individual-level high-stakes ultimatum data that played no part in fitting or selection, offers and rejection rates changed substantially with stake, and stake retained predictive power after conditioning on offer share. The invariance fails.

Under a confirmatory reading this is a negative result. Under the present one it is a boundary measurement, because the contradiction identifies a specific missing quantity, share normalization having removed consequential monetary magnitude. An income-relative monetary coordinate is then a new testable hypothesis, that decisions depend on stake relative to local income rather than on nominal stake. The route runs from geometry to invariance to independent probe to boundary witness to a new representation. A radar return is not a failure. It is information about where the theory stops describing the terrain.

7.4 Symbolic search that reports when it loses

Theory Radar discovers compact classifiers from tabular data and evaluates whether they can fairly replace ensemble models on the dataset at hand (Anonymous, 2026d), combining formula enumeration, search acceleration, representation augmenters, and held-out replay. Its stated value lies in the comparison rather than in symbolic formulas winning, and some datasets admit compact formulas matching or beating ensembles while others do not.

The methodological relevance is that it operationalizes a preference for low-complexity explanation without treating simplicity as sacred, since a formula is accepted only if it survives fair held-out evaluation. Where the boundary is intrinsically high-dimensional the ensemble wins, and that negative result is informative because it identifies a domain in which compression to the searched grammar loses predictive structure. The natural extension scores candidates for stability under admissible representation changes as well as for accuracy and syntactic size, which turns symbolic search into a way of finding simple formulas whose content survives transformations that should not matter.

7.5 Nine registrations and one law

The cases above each exercise part of the loop. The following line exercises all of it, nine times, under sealed registrations, and is the only case on record in which the protocol ran to its declared end (Anonymous, 2026b). It also supplies what the others do not, a measurement of what the protocol costs. Three of the nine registrations failed.

A hub is a row of a corpus appearing in many other rows' nearest-neighbour lists. The instrument counts, for every row, how often it is retrieved among the ten nearest neighbours of the others. The Poisson hub excess is the busiest row's count divided by the largest count a random assignment of the same retrieval slots would reach, so an excess of one is chance. The observer is a linear read, a rotation followed by a diagonal rescaling whose spectrum falls as a power of the index, the exponent running from zero, a pure rotation, to two, a reader that sees one direction. Earlier gates had shown that a corpus's hubs are made by the reader as much as by the corpus. The question was whether the amount of hubness is lawful in the reader's own variables.

The family of laws was declared once and kept. A law is linear in at most three features drawn from a fixed pool of transforms of variables measured on the observed data. The search fits every model of the family on training worlds and ranks by error on held-out worlds under unseen observer spectra, the pilot freezes the best, the registration is sealed by renaming the file with the frozen law inside it, the run uses a fresh seed, and a sealed grader grades. Every bar is a multiple of the frozen law's own pooled pilot error, fixed before the seal, and a competitor beating the law pooled on the transfer groups is a failure by a clause written in advance.

The first four registrations changed the discovery set and kept everything else, and the law changed with the set each time, which is the finding. A law found on Gaussian clouds read the distance concentration and the spectral dimension and failed off Gaussians. A law found on seven shape families held every synthetic family and the SIFT descriptors and missed the Wikipedia embeddings, whose intrinsic dimension of 8 to 20 sat under a spectral dimension of up to 320. The intrinsic dimension was in the pool and the search had dropped it, because on full-dimensional families it duplicates the spectral dimension. Putting manifold worlds into the set made the search take it, and the Wikipedia error fell from 5.3 to 2.5 to 1.0 across the three gates. The discovery set decides the law.

The next two attacked the family rather than the set and both found the same limit. A pool extended with four scale-free tail measures of the neighbour distances was offered to the search, and the search declined every one of them, an outcome the registration had named in advance as its first falsification condition. Placing the heavy tails in the discovery set at the dimensions the transfer asked about then bought the heavy tails and lost the high-dimensional clouds, the previously frozen law was the better predictor pooled, and the gate failed by its own clause. Three registrations had by then put the same worlds on both sides of one limit, the largest error of every law sitting where the hubness skewness exceeds 12, where a law linear in any transform of the skewness cannot turn over. The record said the family, and not the set, was the limit.

The seventh changed the target and nothing else, predicting the log of the Poisson hub excess in place of the skewness. On the same 73 worlds and readers the search froze a law in the concentration of the tenth-neighbour distance, the spectral dimension, and the intrinsic dimension, with pooled pilot error 0.328, a factor of 1.39 in the excess, against 1.135 for the nominal formula and 1.598 for chance. On the run it held the heavy tails, the Gaussian at 512 dimensions, four fresh real slices, and the largest samples at once, and it beat the nominal formula and chance pooled and inside every group, the first law of the line to do so. Same family, same pool, same search, same worlds. The target had been the limit all along.

The eighth asked whether the new target takes coverage without paying for it, and it does

not. Adding the highest-dimensional clouds to the training set bought every full-dimensional world beyond 256 dimensions and the best real-slice errors of the line, and paid on every heavy-tailed world and on an unseen ball, at worse than chance. The earlier law won pooled and the gate failed. The trade the skewness family had shown reappeared in the excess family at about a third of the size and with the same sign.

Before a ninth, an exploration on the rows already collected asked whether any shape of the declared family holds both sides at once. Twelve configurations were scored against the frozen law and none passed. The exploration named the mechanism instead. Under an isotropic reader a uniform ball at 384 dimensions has a tenth-neighbour concentration of 0.003, six times below a Gaussian's at the same dimension, the law's concentration term is a reciprocal, it diverges there, and it predicts an excess near zero against a truth of 1.4. Removing the reciprocal loses the real corpora, which the reciprocal is what fits. That is a boundary of the family's shape and not a missing world, and it is Definition 4 carried out on a declared complexity ordering.

The ninth registered that boundary as the claim and searched nothing. A row is in scope when its tenth-neighbour concentration is at least 0.015, a threshold declared from the collected rows before the pilot and not moved after it. Inside the scope the frozen law is claimed within the earlier bars on fresh worlds. Outside it the law is not claimed, and its pooled error there must exceed the in-scope limit, or the scope was a convenience and the claim fails as stated. That second clause is what stops a scope from being a way of dropping the worlds that fail. The registration also named in advance the one in-scope cell it expected to lose, a heavy-tailed family at a sample size of 12000, with the reason, which is that the law carries no term in sample size.

The run held every bar but that one. All eighteen unseen worlds sat inside the limit on their in-scope rows, pooled 0.31 against a limit of 0.48, the real slices read 0.12 to 0.41, and the law beat both competitors in every group at 0.34 against 1.63 and 1.64. The 180 out-of-scope rows erred at 1.19 against 0.64, which is the second clause holding. The named cell read 0.67 against 0.64. The verdict was indeterminate, by the named cell and no other.

What stands is one law with one declared scope and two recorded boundaries. For a corpus read through a linear observer, on rows whose tenth-neighbour distances vary by at least 1.5 percent, the log of the busiest row's retrieval count over the Poisson ceiling is a linear function of the reciprocal of that variation, the log of the spectral dimension, and the square root of the intrinsic dimension of the observed data, to a factor of about 1.4 in the excess, on every family of world the line built, on corpus slices it never saw, and at four times its discovery sample. Below 1.5 percent variation the concentration term diverges and the law is withdrawn. For one heavy-tailed family the law under-predicts by about a factor of three under the isotropic reader, and a sweep after the line closed showed this to be a step in sample size between 4000 and 8000 followed by a plateau, which a law with no term in sample size cannot follow. Neither boundary is repaired by a change of the discovery set.

Three properties are visible in the record rather than asserted. The bars never moved, and three registrations failed on clauses their own authors had written. Two pilots froze a law that trailed a competitor on the pilot's own rows, and both were sealed and run as registered with that fact recorded, because the transfer groups are the test and the pilot is not. And when a registration that would pass was wanted, the only route the record allowed was a change of claim, a boundary declared before the run with its risk named, and the run returned the outcome the risk named. The alternative route, dropping the worlds that fail without calling that a scope, was available and was not taken.

The sequence of causes is the finding, and none of its five statements was available at the first registration. The discovery set decides the law. The family cannot hold both sides of the divide. The target was the limit. Coverage moves the error under the new target too. The boundary is a

divergent term and not a missing world. Each was forced by a failure that had been reduced rather than absorbed.

7.6 The check found a defect its own authors had not

The preceding cases share a weakness a reader should hold against them. Every one is the work of the people proposing the discipline, so each is a report of a method applied by its authors to their own theories. The following episode is the nearest thing to independent evidence the programme can currently offer, because in it the method contradicted the people running it.

The four record types the protocol needs were implemented as machine-readable files, among them a transformation registry declaring, per track, the admissible families with the date each was declared. A checker was then written from Definition 1, testing that every registered test names a family the registry carries. Run against the programme’s own registries for the first time, it reported ten tests, entered across a single day of gate records, citing three families the registry did not contain.

The finding was bookkeeping rather than a priority violation, and establishing which took evidence rather than memory. Each of the three families was declared in its own gate’s sealed registration before the corresponding run, which is checkable in two ways that do not rely on anybody’s recollection. Each seal commit is an ancestor of the commit that added that gate’s results, and each sealed file still hashes to the value the track document recorded. The declarations existed and preceded their tests, and only the registry was late.

Two things about the episode bear on the argument. The first is that the defect had been visible for two days in a published artifact and had not been seen. The generator rendering each entry’s invariance envelope prints a transformation identifier followed by the sentence describing the family, so an undeclared identifier printed an empty sentence, thirty times across four entries and the pages built from them. Reading had not caught it. A check derived from a definition in this paper caught it on first execution.

The second is what the repair had to look like. Adding the three families with the current date would have made the registry pass and would have misdated the declarations. Adding them with the declaration date alone would have concealed that the registry was late. Deleting the offending tests would have removed the record of tests that ran, which is the file drawer the discipline exists to prevent. The rows were therefore entered carrying both dates and a pointer to the sealed artifact, so the registry now says, in a form a stranger can check, that these three declarations were made on one date and recorded on another. The specification was amended to require exactly this, and to state the condition under which a late row is a repair rather than a forgery, which is that the declaration it points at can be shown to precede the result.

We report this because it is the only case here in which the discipline cost its authors something, and because it illustrates the narrow claim of Section 2. No new criterion of invariance was involved. What found the defect was the requirement that the admissible class be a declared artifact rather than a sentence, and what made the repair something other than cosmetic was the requirement that the record show its own history.

8 The fundamental structure conjecture

The cases motivate a conjecture rather than a conclusion.

Conjecture 1 (Fundamental Structure Conjecture). *Many apparently high-dimensional and domain-specific phenomena admit low-complexity invariant structures acting on richer domain-specific rep-*

representations, and those structures can be progressively isolated by searching for what survives admissible changes of representation, scale, population, and task.

The conjecture is broad and therefore easy to overstate, and three qualifications are essential. Low dimensionality is not guaranteed, and some phenomena may be irreducibly complex at the scale of interest, the cases in which ensembles beat compact formulas being exactly the evidence a serious programme preserves, along with the shape boundary of Section 7.5 where the declared family had no form holding both sides. Invariance is always relative to a transformation class, so what survives translation may fail under a change of scale and what survives within a culture may fail across cultures. And invariance does not imply ontology, being evidence that a structure depends less on chosen descriptions rather than proof that it is metaphysically ultimate.

The conjecture nonetheless supplies a search strategy. Begin with a rich representation, identify a compact structure, transform the representation, transfer the structure, and keep what survives. Repeated quotienting of representational contingencies can expose an empirical core that becomes progressively harder to remove without losing predictive power.

9 Elegance, rigour, and the risk of beautiful error

The discipline gives a precise role to two virtues. Elegance pushes toward fewer primitives, reusable structures, and compression across apparently unrelated phenomena. Rigour prevents elegance from becoming an aesthetic substitute for evidence.

The danger is plain. A mathematically beautiful model can be wrong, and the more beautiful it is the easier it is to rationalize away contrary evidence. Generating hostile tests therefore belongs to the theory rather than to a later optional exercise, and a mature research artifact should carry a claim ledger, a separation of definitions from measured claims, a registry of admissible transformations, preregistered gates, holdout data, ablations, and explicit statements of what would force revision. Section 7.5 shows what the last of those costs when it is enforced, which is three failed registrations out of nine.

This is consistent with a pragmatist stance. The reality of a proposed structure is not secured by metaphysical assertion. It earns standing by generating correct, transferable predictions and by surviving increasingly severe attempts to remove representational artifacts.

10 A research programme

Five workstreams follow.

A calculus of philosophical invariants. Develop a common formal language for equivalence claims across epistemology, decision theory, ethics, philosophy of mind, and philosophy of science. Each application specifies the representation space, the output space, the transformation set, the output equivalence, non-degeneracy criteria, uncertainty sets, and witness complexity. The immediate deliverable is a taxonomy of invariance claims, which are exact, approximate, equivariant, conditional, local, distributional, and scale-dependent.

Representation attacks. Construct benchmarks in which the same underlying cases are presented through alternative coordinate systems, embeddings, languages, schemas, and modalities, requiring candidate theories to survive invertible transformations and to degrade predictably under lossy ones. This connects the discipline to representation learning and symmetry discovery.

Low-rank evaluation. Test whether evaluator geometry inferred in one domain transfers to another under independently determined representations. The decisive experiments identify an

evaluator in one domain, freeze the structure, and predict choices in two others with minimal target-specific adjustment.

Automated theory search. Extend symbolic search from classification toward general structure discovery with explicit complexity and invariance objectives, with candidate outputs including equations, metric tensors, low-rank subspaces, transformation groups, and boundary conditions, and with discovered structures replayed on untouched test data.

Boundary atlases. Rather than publishing only the domains where a theory works, build atlases of where invariances fail, estimating for each claimed symmetry the boundary in scale, context, population, or representation at which the effect becomes detectable. These maps may be worth more than a list of successes, because they name the variables the next theory needs. The two boundaries of Section 7.5, one in the concentration of a distance and one in sample size, are entries in such an atlas, and each names a term a successor would have to carry.

11 Objections

This is falsificationism with software. The discipline is compatible with falsificationist virtues and is not exhausted by binary refutation. Its distinctive object is the structure of the counterexample, and minimal witnesses, transformation grammars, and boundary estimates turn a failed prediction into information about which claimed equivalence stopped holding. The method is closer to debugging a formal architecture than to recording a rejected hypothesis (Zeller and Hildebrandt, 2002).

Invariance is not truth. Correct. A false theory can be invariant and a true local theory can legitimately fail outside its domain. van Fraassen (1989) presses the stronger version, that symmetry arguments carry substantive commitments about what counts as a difference, and the present account accepts it by treating the transformation class as declared and contestable rather than discovered. Invariance is therefore comparative evidence against representational arbitrariness and not a truth criterion, and predictive adequacy, independent replication, and domain specification remain necessary.

The transformation class is chosen by the theorist. Also correct, and it is the central governance problem of any invariance-based method, the one Nozick (2001) raises without settling and the one the robustness literature treats as its main risk. The remedy is not to pretend the choice is objective. It is to make it explicit, versioned, contestable, and empirically revisable, so competing philosophical positions appear as competing transformation registries and the disagreement is located rather than argued. In the programmes reported here the registry is a file, it is sealed before the runs that test it, and Section 7.5 is a case in which the declared family and not the data turned out to be the limit.

Philosophical claims are often normative. Normative claims are not settled by observing behaviour, and the discipline does not infer an ought from an is. It tests the operational consequences of a normative principle once the principle is formalized, which means whether an implementation respects declared equivalences, whether a proposed standard is internally invariant, whether a policy rule produces representation-dependent outcomes, and whether a normative model predicts human judgments when offered as a descriptive theory.

Search will overfit. Automated search creates severe multiplicity and researcher degrees of freedom, so the discipline requires separation of discovery from evaluation, fair holdout replay, preregistration where feasible, and independent replication. An expression found after searching millions of candidates is not evidence until its search exposure is accounted for and its performance survives untouched data. Section 7.5 is the worked case, the search having run nine times over

one declared family with every bar a multiple of a pilot error fixed before the seal, and the two gates whose frozen law trailed a competitor on the pilot's own rows run as registered rather than re-searched.

The evidence is the proponents' own work. This is the objection we think strongest. Every case in Section 7 comes from one programme, and a method applied by its authors to their own theories is not independently tested, however sealed its registrations. Three things mitigate and none removes it. The registrations, data, graders, and run logs are public, so the verdicts are checkable even though they are not independent. The record retains outcomes that embarrass the programme, among them three failed registrations, two indeterminate gates, a published invariance that failed on external data, and a population in which a deterministic theorem holds for half the respondents. And Section 7.6 is a case in which the discipline's own check overruled its authors. What is missing is a second programme, run by other people, and until that exists the claims here are claims about a method demonstrated once.

12 Engineering as an epistemic instrument

The phrase philosophy engineering can suggest that philosophy supplies requirements and engineering implements them. The relationship proposed here is stronger. Engineering exposes the hidden commitments of a philosophy by forcing them into interfaces, data structures, registries, test cases, and failure modes, and ambiguities surviving prose become hard to ignore when a system must decide whether two cases belong to the same equivalence class.

This changes the epistemic role of implementation. A program is not only a demonstration that a theory can be instantiated. It can be a measurement instrument for the theory. Canonicalizers test claims about sameness. Equivariant architectures test claims about symmetry. Symbolic search tests claims about compressibility. Cross-domain transfer tests claims about generality. Minimal witnesses test claims about irrelevance. Boundary maps test claims about scope.

The resulting practice is neither armchair philosophy nor ordinary empirical science. It is a disciplined traffic between conceptual analysis and experiment. Philosophy proposes distinctions, mathematics makes them precise, engineering makes them executable, experiment pressures them, and philosophy must then explain the returns.

The operational sense of fundamental that follows is modest. Whatever survives repeated removal of representational contingencies, while preserving discriminative and predictive adequacy, is a better candidate for fundamental structure than what disappears under harmless redescription. The search can fail, and may reveal no compact invariant core. That too is a substantive result.

13 Conclusion

The idea that what survives admissible transformation is a better candidate for the real has been stated four times already, in geometry, in robustness analysis, in the theory of causal explanation, and in structural realism. This paper does not improve on those statements. It argues that the idea has been held in a form that cannot be audited, and that the three things a prose statement leaves in the claimant's hands, the admissible class, the reduction of the failure, and the revision, can be made into declared artifacts a checker refuses when they are absent.

What follows is a discipline whose methodology is six verbs. Formalize, derive, search, probe, witness, revise. We have reported it running on four graded predictions of one theory, on a published invariance that failed on independent data, and on a discovery line that asked one question nine times and ended with one law, one declared scope, and two boundaries. We have also reported the

case in which the discipline’s own check found a defect in the records of the people who wrote it, which is the only evidence here not resting on our good faith.

The deepest promise is not that philosophy is replaced by engineering, nor that every profound question has a computational answer. It is that some philosophical propositions can be made precise enough to become experimentally vulnerable. Once that happens, implementation stops being application and becomes inquiry.

The guiding question is therefore not whether the fundamental theory has been found. It is more disciplined. What remains after every distinction that should not matter has been transformed away, every compact structure has been attacked on unseen data, and every failure has been reduced to the smallest witness available? Whatever survives that has earned the right to be taken seriously as fundamental.

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